Week 9: Markov Chain Monte Carlo Metropolis–Hastings and Gibbs Sampling

Sabyasachi Chatterjee

STAT 542: Statistical Learning
University of Illinois Urbana-Champaign
Fall 2026

Contents

1 Introduction to Bayesian Computation **3**

2 The MCMC Principle **3**

3 Markov Chains **4**
3.1 Stationary Distributions 4
3.2 Ergodic Theorem 4

4 An Illustrative Example **5**
4.1 A Markov Chain on the State Space 5
4.2 Why Does This Work? 5
4.3 Symmetry and the Uniform Distribution 6
4.4 A New Perspective 6

5 Detailed Balance **7**
5.1 A Graphical Interpretation 7
5.2 An Example 7

6 The Metropolis–Hastings Algorithm **8**
6.1 Deriving the Acceptance Probability 8
6.2 Implementing the Algorithm 8
6.3 Why the Algorithm Works 8
6.4 Why the Normalizing Constant is Unnecessary 9

7 Random Walk Metropolis: An Example and Proposal Choice **9**
7.1 Example: Sampling from a Power-Law Distribution 10
7.2 Proposal Choice and the Price of Dependence 10

8 Gibbs Sampling **10**
8.1 The Algorithm 11
8.2 Gibbs Sampling as a Special Case of Metropolis–Hastings 11
8.2.1 Metropolis-within-Gibbs 12
8.3 Practical Considerations 12

8.4 Examples of the Gibbs Sampler 13
8.4.1 Bivariate Normal 13
8.4.2 Linear Regression 14
8.4.3 Probit Regression: Why Introduce Latent Variables? 14
8.5 Gibbs Sampler for Probit Regression 14
8.5.1 The Two Conditional Distributions 15
8.5.2 Simulating the Truncated Normals 15
8.5.3 The Gibbs Algorithm 16

1 Introduction to Bayesian Computation

In Bayesian inference, our goal is to compute posterior summaries of unknown parameters. Let θ denote the unknown parameter and $\pi(\theta)$ its posterior density given the observed data. Typical quantities of interest include posterior means, variances, probabilities, credible intervals and posterior predictive distributions. Many of these can be expressed in terms of posterior expectations:

$$E_\pi[g(\theta)] = \int g(\theta)\pi(\theta)\, d\theta. \tag{1}$$

Unfortunately, posterior distributions are often analytically intractable. Bayes' theorem gives

$$\pi(\theta) = \frac{L(\theta)p(\theta)}{\int L(\vartheta)p(\vartheta)\, d\vartheta},$$

where L is the likelihood and p the prior density. The denominator, called the *marginal likelihood* or *evidence*, is often impossible to evaluate analytically and computationally prohibitive even numerically. Many Bayesian procedures, however, can be implemented without evaluating this normalizing constant.

Two major approaches are *Markov Chain Monte Carlo* (MCMC) and *variational inference* (VI). MCMC constructs a Markov chain whose stationary distribution is the posterior; running it for sufficiently long gives approximate posterior samples. VI instead approximates the posterior by a simpler distribution chosen through optimization. It is often faster but introduces approximation error. In these notes we begin with MCMC.

2 The MCMC Principle

Suppose our target is a probability distribution π on a state space S, usually the posterior distribution. If we could generate independent samples $\theta^{(1)}, \ldots, \theta^{(N)} \overset{\text{i.i.d.}}{\sim} \pi$, then the ordinary law of large numbers would give

$$\frac{1}{N}\sum_{t=1}^{N} g(\theta^{(t)}) \longrightarrow \int g(\theta)\pi(\theta)\, d\theta \tag{2}$$

almost surely, provided g is integrable.

The difficulty is that independent sampling from π is typically infeasible. The key idea behind MCMC is to generate a dependent sequence $\theta^{(0)}, \theta^{(1)}, \ldots$ forming a Markov chain with stationary distribution π. Under suitable conditions, time averages still converge to posterior expectations, allowing the approximation

$$\frac{1}{N}\sum_{t=1}^{N} g(\theta^{(t)}) \approx \int g(\theta)\pi(\theta)\, d\theta. \tag{3}$$

Most MCMC algorithms require only quantities proportional to $\pi(\theta)$, precisely what Bayes' theorem usually provides.

This leaves two questions: how do we construct a chain with the desired stationary distribution, and why do its averages converge to expectations under π? We first review the basic concepts of Markov chains.

3 Markov Chains

A sequence $\Theta^{(0)}, \Theta^{(1)}, \ldots$ taking values in S is a *Markov chain* if, conditional on the present state, the future is independent of the past. More precisely, for every measurable set A,

$$P\Big(\Theta^{(t+1)} \in A \mid \Theta^{(t)}, \ldots, \Theta^{(0)}\Big) = P\Big(\Theta^{(t+1)} \in A \mid \Theta^{(t)}\Big). \tag{4}$$

Thus the current state contains all the information required to generate the next state. We consider only *time-homogeneous* chains, whose transition rule does not depend on time.

For a finite or countable state space, the transition probabilities are

$$P(x, y) = P\Big(\Theta^{(t+1)} = y \mid \Theta^{(t)} = x\Big). \tag{5}$$

They satisfy $P(x, y) \geq 0$ and $\sum_y P(x, y) = 1$. On a continuous state space, *when a transition density exists*, we similarly write $P(\Theta^{(t+1)} \in A \mid \Theta^{(t)} = x) = \int_A P(x, y)\, dy$, with $P(x, y) \geq 0$ and $\int P(x, y)\, dy = 1$. More generally, the transition kernel is written $P(x, A)$; this also allows a positive probability of remaining exactly at x, as occurs when an MCMC proposal is rejected.

3.1 Stationary Distributions

A probability distribution π is *stationary* (or *invariant*) if starting the chain from π leaves its distribution unchanged after one step: $\Theta^{(t)} \sim \pi$ implies $\Theta^{(t+1)} \sim \pi$. For a finite or countable state space, this means

$$\pi(y) = \sum_{x \in S} \pi(x) P(x, y), \qquad y \in S. \tag{6}$$

When the relevant densities exist, the continuous version is

$$\pi(y) = \int_S \pi(x) P(x, y)\, dx. \tag{7}$$

The stationary distribution is the equilibrium distribution: if $\Theta^{(0)} \sim \pi$, every subsequent iterate has distribution π, although the iterates need not be independent.

3.2 Ergodic Theorem

Stationarity alone does not guarantee that a chain started elsewhere will explore the target distribution. On a finite or countable state space, a chain is *irreducible* if every state can be reached from every other state with positive probability in some finite number of transitions. The analogous conditions on continuous state spaces are more technical.

Theorem 3.1 (Ergodic Theorem)**.** *Suppose $\{\Theta^{(t)}\}$ is an irreducible Markov chain on a finite or countable state space, with stationary distribution π. Then, for every integrable function g, from any initial state,*

$$\frac{1}{N} \sum_{t=1}^{N} g(\Theta^{(t)}) \longrightarrow E_\pi[g(\Theta)] \quad \textit{almost surely.}$$

The same conclusion holds on a general state space under positive Harris recurrence, the recurrence condition used for general-state-space MCMC.

This is the Markov-chain counterpart of the law of large numbers: posterior expectations can be approximated by empirical averages along a single trajectory, even though successive samples are dependent. Convergence of the *distribution of the current state* to π is a separate conclusion; for an irreducible finite chain, it additionally requires *aperiodicity*. Aperiodicity is not needed for the time-average theorem above.

4 An Illustrative Example

Suppose we wish to sample uniformly from binary sequences of length d having no two consecutive ones. For $d = 5$, the sequences 00000, 01010 and 10100 are allowed, whereas 11000 and 01110 are not. Formally,

$$S = \Big\{ x = (x_1, \ldots, x_d) \in \{0,1\}^d : x_i x_{i+1} = 0 \text{ for } i = 1, \ldots, d-1 \Big\}.$$

One possible approach is rejection sampling: generate each bit independently with probability $1/2$ of being one, and reject the sequence if it contains consecutive ones. Unfortunately, this quickly becomes impractical. The number of valid sequences is $N_d = F_{d+2}$, where F_k is the kth Fibonacci number. Since $F_k \sim \varphi^k/\sqrt{5}$ with $\varphi = (1+\sqrt{5})/2 \approx 1.618$, the acceptance probability is

$$\frac{N_d}{2^d} \sim \frac{\varphi^2}{\sqrt{5}} \left(\frac{\varphi}{2}\right)^d.$$

Here $\varphi/2 \approx 0.809$, so acceptance decreases exponentially with d; for $d = 100$ it is about 7.3×10^{-10}. We therefore need a different strategy.

4.1 A Markov Chain on the State Space

Instead of generating an entire sequence from scratch, suppose we already have a valid sequence $X = (X_1, \ldots, X_d) \in S$ and make a small random modification:

1. Choose a coordinate $i \in \{1, \ldots, d\}$ uniformly at random.
2. Propose flipping that bit, $X_i \longrightarrow 1 - X_i$.
3. Accept if the resulting sequence belongs to S; otherwise reject the proposal and remain at the current state.

Each transition changes at most one coordinate, so successive states are very similar. Such local moves are typical in MCMC.

For example, from $X = (1, 0, 0, 1, 0)$, flipping coordinate 2, 3 or 5 creates consecutive ones and is rejected. Flipping coordinate 1 gives $(0, 0, 0, 1, 0)$ and is accepted. Thus the constraint determines which proposals are accepted.

4.2 Why Does This Work?

Can every valid sequence be reached, or might the chain become trapped? We first check irreducibility and aperiodicity, then identify the stationary distribution.

Proposition 4.1. *The Markov chain described above is irreducible.*

Proof. Take $x, y \in S$. Starting from x, flip every one to zero. These moves cannot create consecutive ones, so all intermediate states are valid and we reach $(0, \ldots, 0)$. Now flip precisely those coordinates where y has value one. Since y is valid, every intermediate state is again valid. Each move has positive probability, giving a path from x to y. □

Proposition 4.2. *For $d \geq 2$, the Markov chain is aperiodic.*

Proof. A rejected proposal leaves the chain at its current state, giving a self-loop. For example, $(1, 0, 0, \ldots, 0)$ rejects a flip of its second coordinate. In an irreducible chain all states have the same period, and a self-loop makes this period one. For $d = 1$, one may add a holding probability to make the chain aperiodic. □

The chain is therefore irreducible and aperiodic (for $d \geq 2$). It remains to identify its stationary distribution.

4.3 Symmetry and the Uniform Distribution

The key observation is that the transition rule is symmetric. If two distinct valid states x, y differ in exactly one coordinate, then $P(x, y) = P(y, x) = 1/d$: flipping that coordinate takes either state to the other. If they differ in more than one coordinate, both probabilities are zero. Thus

$$P(x, y) = P(y, x) \tag{8}$$

for every pair of states.

For the uniform distribution $\pi(x) = 1/|S|$, symmetry gives

$$\pi(x)P(x, y) = \frac{1}{|S|}P(x, y) = \frac{1}{|S|}P(y, x) = \pi(y)P(y, x).$$

The probability flowing from x to y equals that flowing back from y to x. Summing over x gives $\sum_x \pi(x)P(x, y) = \pi(y) \sum_x P(y, x) = \pi(y)$, proving stationarity. The ergodic theorem therefore yields

$$\frac{1}{N} \sum_{t=1}^{N} g(X^{(t)}) \longrightarrow \frac{1}{|S|} \sum_{x \in S} g(x) \quad \text{almost surely.}$$

Moreover, irreducibility and aperiodicity imply that the current state is approximately uniform after sufficiently many steps.

4.4 A New Perspective

The example illustrates the MCMC philosophy: construct local moves that preserve the target distribution, then use a long trajectory to estimate expectations. Symmetry made the uniform target easy to obtain. For an arbitrary target π, we need a more general principle for designing transitions. The pairwise probability-flow calculation above suggests just such a principle: *detailed balance*.

5 Detailed Balance

Imagine many copies of the chain whose current states have distribution π. The fraction moving from x to y in one transition is $\pi(x)P(x,y)$, while $\pi(y)P(y,x)$ moves in the reverse direction. Equality means that there is no net probability flow between the two states, just as in the binary-sequence example.

Definition 5.1. *A Markov chain satisfies* detailed balance *with respect to π if*

$$\pi(x)P(x,y) = \pi(y)P(y,x) \tag{9}$$

for every pair of states $x, y \in S$.

This is also called *local balance*, because it balances each pair individually. The displayed condition applies to discrete chains and, almost everywhere, to continuous chains with densities; for general kernels it means that $\pi(dx)P(x,dy)$ is symmetric in x, y.

Theorem 5.2. *If a Markov chain satisfies detailed balance with respect to π, then π is a stationary distribution.*

Proof. For every state y,

$$\sum_{x\in S} \pi(x)P(x,y) = \sum_{x\in S} \pi(y)P(y,x) = \pi(y)\sum_{x\in S} P(y,x) = \pi(y),$$

which is the stationarity condition. With densities, replace sums by integrals; the same argument applies to general kernels. □

Detailed balance is sufficient, but not necessary, for stationarity. It is useful because pairwise identities are often easy to verify and lead to simple constructions of chains with a prescribed target.

5.1 A Graphical Interpretation

Represent states by vertices of a graph, with transitions along its edges. Think of probability flowing through pipes between the vertices. Under detailed balance, every pipe carries equal amounts in the two directions, so there is no net accumulation or depletion at any vertex and the distribution remains unchanged. Stationarity alone only requires total inflow to equal total outflow at each vertex; circulating currents can still occur.

5.2 An Example

Consider a random walk on a finite connected undirected graph with vertices $S = \{1, \dots, n\}$, $n \geq 2$. Let d_i be the degree of vertex i. From i, choose a neighbour uniformly, so

$$P(i,j) = \frac{1}{d_i} I\{i \sim j\}, \qquad \pi(i) = \frac{d_i}{\sum_{k=1}^{n} d_k},$$

where $i \sim j$ denotes adjacency. Then

$$\pi(i)P(i,j) = \frac{I\{i \sim j\}}{\sum_k d_k} = \pi(j)P(j,i).$$

Thus detailed balance holds, and the stationary distribution is proportional to vertex degree. Vertices with many neighbours are visited more frequently than vertices with few neighbours.

6 The Metropolis–Hastings Algorithm

In the examples so far, we chose simple transitions and then identified their stationary distribution. In Bayesian statistics the task is reversed: the target π is fixed in advance, and we must construct a chain having π as its stationary distribution. Metropolis–Hastings provides a general solution: propose a move using a convenient distribution, then occasionally reject it to restore detailed balance.

6.1 Deriving the Acceptance Probability

Suppose the current state is x and we generate a proposal $Y \sim Q(x, \cdot)$. The proposal distribution is chosen by us and need not preserve π: in general, $\pi(x)Q(x, y) \neq \pi(y)Q(y, x)$. The key idea is to reduce whichever of these two probability flows is larger until they agree.

Let $A(x, y)$ be the probability of accepting a proposal from x to y. If it is rejected, the chain remains at x. For distinct states,

$$P(x, y) = Q(x, y)A(x, y), \qquad x \neq y, \tag{10}$$

so detailed balance requires

$$\pi(x)Q(x, y)A(x, y) = \pi(y)Q(y, x)A(y, x). \tag{11}$$

If $\pi(x)Q(x, y) > \pi(y)Q(y, x)$, there is too much flow from x to y. We accept every reverse move, setting $A(y, x) = 1$. Equation (11) then fixes the forward acceptance probability as the ratio of reverse to forward flow. If the inequality is reversed, we instead accept every forward move and reduce the reverse flow. Combining the cases gives

$$A(x, y) = \min\left\{1, \frac{\pi(y)Q(y, x)}{\pi(x)Q(x, y)}\right\}. \tag{12}$$

Thus the acceptance rule has a simple interpretation: keep the smaller flow unchanged and reduce the larger one to match it. This accepts as many moves as possible while balancing the two directions. A proposal outside the support of π is rejected; so is a proposed move with positive forward flow but zero reverse flow.

6.2 Implementing the Algorithm

Initialize at a state $\theta^{(0)}$ with positive target density. At each iteration $t = 0, 1, \ldots$, let $x = \theta^{(t)}$ and:

1. Generate a proposal $y \sim Q(x, \cdot)$.
2. Compute $A(x, y)$ in (12) and independently generate $U \sim \text{Uniform}(0, 1)$.
3. Set $\theta^{(t+1)} = y$ if $U \leq A(x, y)$; otherwise set $\theta^{(t+1)} = x$.

Rejected moves are part of the chain: the repeated current state must be retained when computing empirical averages.

6.3 Why the Algorithm Works

Theorem 6.1. *The Metropolis–Hastings chain satisfies detailed balance with respect to π, and hence has stationary distribution π.*

Proof. For $x \neq y$, substituting the acceptance probability gives

$$\pi(x)P(x,y) = \min\{\pi(x)Q(x,y), \pi(y)Q(y,x)\} = \pi(y)P(y,x). \tag{13}$$

The middle expression is symmetric in x, y, and the identity for $x = y$ is automatic. Thus detailed balance holds and implies stationarity. For continuous proposals, the same calculation balances the off-diagonal transition densities; the probability of staying at the current state also satisfies detailed balance automatically. □

Stationarity does not by itself guarantee exploration from an arbitrary starting point. The proposal must also allow the chain to explore the target support, under the conditions discussed in Section 3.

6.4 Why the Normalizing Constant is Unnecessary

Suppose $\pi(\theta) = \widetilde{\pi}(\theta)/Z$, where $Z = \int \widetilde{\pi}(\theta)\, d\theta$ is unknown. In the acceptance ratio,

$$\frac{\pi(y)Q(y,x)}{\pi(x)Q(x,y)} = \frac{\widetilde{\pi}(y)Q(y,x)}{\widetilde{\pi}(x)Q(x,y)},$$

so Z cancels. This is precisely what we need in Bayesian inference: Bayes' theorem gives $\pi(\theta \mid \text{data}) \propto L(\theta)p(\theta)$, and we can run the algorithm without computing the marginal likelihood.

The proposal is flexible: it may be Gaussian, uniform, a random walk, or another distribution depending on the current state. We need to be able to simulate it and evaluate the forward and reverse proposal probabilities or densities. The acceptance step preserves the same target for all such choices, although their efficiencies can differ greatly.

7 Random Walk Metropolis: An Example and Proposal Choice

One of the simplest proposals makes a small random perturbation of the current state: $Y = x + Z$, where Z has a distribution symmetric about zero. Examples include $Z \sim N(0, \Sigma)$ and a uniform distribution on $[-a_1, a_1] \times \cdots \times [-a_d, a_d]$. Since $f_Z(z) = f_Z(-z)$,

$$Q(x,y) = f_Z(y - x) = f_Z(x - y) = Q(y,x). \tag{14}$$

The proposal terms therefore cancel from the acceptance ratio, giving

$$A(x,y) = \min\left\{1, \frac{\pi(y)}{\pi(x)}\right\}. \tag{15}$$

With a symmetric proposal this is called the *Metropolis algorithm*. Propose $Y = x + Z$ and accept according to (15); on rejection, remain at x.

Moves to a higher-density region are always accepted. Moves to a lower-density region are accepted with probability $\pi(y)/\pi(x)$. Thus the chain favours regions where the target density is large while still exploring lower-density regions. These occasional downhill moves are crucial: without them, the chain could become trapped at local modes.

7.1 Example: Sampling from a Power-Law Distribution

Let $S = \{1, \ldots, M\}$, with $M \geq 2$, and consider the probability mass function

$$\pi(i) = \frac{i^{-\alpha}}{\sum_{k=1}^{M} k^{-\alpha}}, \qquad \alpha > 0. \tag{16}$$

Power laws arise in applications including network science, linguistics, genetics and statistical physics. The normalizing sum may be expensive when M is large, but Metropolis sampling only needs ratios.

From state i, propose $i + 1$ or $i - 1$, each with probability $1/2$. If the proposed value lies outside S, stay at i. This endpoint rule keeps the proposal symmetric: $Q(i, j) = Q(j, i) = 1/2$ for neighbouring states. For a valid proposal j, accept with probability

$$A(i, j) = \min\left\{1, \frac{\pi(j)}{\pi(i)}\right\} = \min\left\{1, \left(\frac{i}{j}\right)^{\alpha}\right\}. \tag{17}$$

For example, assuming $M \geq 21$, a proposal from 20 to 21 is accepted with probability $(20/21)^{\alpha}$, whereas a proposal from 20 to 19 is always accepted. The chain therefore favours smaller integers, where the target probability is higher, while still moving to larger ones. The normalizing constant never enters the calculation.

7.2 Proposal Choice and the Price of Dependence

Although a symmetric proposal disappears from the acceptance ratio, it has a profound effect on efficiency. With a very small proposal variance, almost every move is accepted, but the chain takes tiny steps and explores slowly. With a very large variance, proposals often land in low-density regions and are rejected, so the chain again moves slowly. An effective proposal balances the size of accepted moves against their frequency; high acceptance alone does not imply efficient sampling.

This is the promise and the cost of MCMC. Many simple stochastic updates can explore a complicated target without sampling from it directly, but successive iterates are generally dependent. For random walk Metropolis, strong positive autocorrelation typically makes a fixed number of iterations less informative than the same number of independent samples. Careful proposal choice helps the chain mix more rapidly and makes the empirical averages more accurate for a given computational budget.

8 Gibbs Sampling

Metropolis–Hastings can explore the target slowly when many proposals are rejected. Gibbs sampling avoids rejection by drawing each coordinate exactly from its conditional distribution. Suppose the target is $\pi(\theta_1, \ldots, \theta_p)$ and that, although direct joint sampling is difficult, we can sample from each *full conditional* $\pi(\theta_j \mid \theta_{-j})$, where θ_{-j} denotes all coordinates except θ_j. The idea is to update one coordinate at a time while holding the others fixed.

8.1 The Algorithm

Starting from $\theta^{(0)}$, one *systematic-scan* Gibbs iteration (or sweep) updates $j = 1, \dots, p$ in order:

$$\theta_j^{(t+1)} \sim \pi\Big(\theta_j \mid \theta_1^{(t+1)}, \dots, \theta_{j-1}^{(t+1)}, \theta_{j+1}^{(t)}, \dots, \theta_p^{(t)}\Big).$$

Newly sampled values are used immediately in later updates of the same sweep. In two dimensions this simply means alternating between $\theta_1^{(t+1)} \sim \pi(\theta_1 \mid \theta_2^{(t)})$ and $\theta_2^{(t+1)} \sim \pi(\theta_2 \mid \theta_1^{(t+1)})$. Alternatively, a *random-scan* sampler chooses one coordinate at each iteration, with fixed positive selection probabilities, and updates only that coordinate. Coordinates may also be grouped into blocks and sampled jointly from their full conditionals.

Gibbs sampling requires no proposal variance and accepts every update. Once the conditionals have been derived, implementation is often simple; conditionally conjugate models are particularly convenient. However, exact conditional simulation must be possible, and acceptance alone does not ensure rapid exploration: strongly dependent coordinates can lead to slow mixing. When a conditional cannot be sampled exactly, a Metropolis–Hastings step can replace that update, as described below.

8.2 Gibbs Sampling as a Special Case of Metropolis–Hastings

Why is no acceptance–rejection correction needed? Consider an update of the first coordinate from $x = (x_1, x_2)$. We draw $y_1 \sim \pi(\theta_1 \mid x_2)$ and propose $y = (y_1, x_2)$. On this fixed-coordinate slice, the forward and reverse proposal densities are

$$Q(x, y) = \pi(y_1 \mid x_2), \qquad Q(y, x) = \pi(x_1 \mid x_2). \tag{18}$$

For a continuous state space, the second coordinate is held fixed by a point mass; these are densities in the coordinate being updated, rather than ordinary joint densities on the entire state space.

The Metropolis–Hastings acceptance probability is

$$A(x, y) = \min\left\{1, \frac{\pi(y)Q(y, x)}{\pi(x)Q(x, y)}\right\}. \tag{19}$$

Using $\pi(x_1, x_2) = \pi(x_1 \mid x_2)\pi(x_2)$, the ratio becomes

$$\frac{\pi(y_1, x_2)\pi(x_1 \mid x_2)}{\pi(x_1, x_2)\pi(y_1 \mid x_2)} = \frac{\pi(y_1 \mid x_2)\pi(x_2)\pi(x_1 \mid x_2)}{\pi(x_1 \mid x_2)\pi(x_2)\pi(y_1 \mid x_2)} = 1. \tag{20}$$

Thus $A(x, y) = 1$: the proposal already balances the probability flow between the two states. The same calculation applies to any coordinate or block.

Theorem 8.1. *Each Gibbs coordinate or block update is a Metropolis–Hastings update with acceptance probability one. It satisfies detailed balance with respect to π and therefore preserves π.*

Remark 8.1 (Stationarity and detailed balance in Gibbs sampling)**.** *Each coordinate-update kernel K_j satisfies detailed balance with respect to π and therefore preserves π.*

1. *A systematic sweep preserves π because invariance is preserved under composition: $\pi K_1 \cdots K_d = \pi$.*

2. *Although each coordinate update satisfies detailed balance, a full systematic sweep need not. Indeed, reversing the sweep $K_1 \cdots K_d$ gives $K_d \cdots K_1$, which generally differs from the original kernel.*

3. *Random scan satisfies detailed balance. If coordinate j is selected with fixed probability p_j, its kernel is $K = \sum_j p_j K_j$, and*

$$\pi(dx)K(x, dy) = \sum_j p_j \pi(dx)K_j(x, dy) = \sum_j p_j \pi(dy)K_j(y, dx) = \pi(dy)K(y, dx).$$

Stationarity alone does not guarantee exploration or convergence; appropriate irreducibility and recurrence conditions are also needed.

8.2.1 Metropolis-within-Gibbs

If, for example, $\pi(\theta_1 \mid \theta_2)$ is available only up to a normalizing constant, hold θ_2 fixed and perform a Metropolis–Hastings step targeting this conditional. Other coordinates can still use exact Gibbs steps. Such *Metropolis-within-Gibbs* algorithms retain the coordinatewise structure while allowing more general models; the conditional normalizing constant cancels in the acceptance ratio.

8.3 Practical Considerations

Constructing a chain with the right stationary distribution is the first step. We must also assess whether a finite run has explored that distribution adequately.

Burn-in. An arbitrary initial state may lie far from typical target values, so early iterates need not resemble target samples. Under suitable ergodicity conditions (irreducibility and aperiodicity suffice for a finite chain with a stationary distribution), the current-state distribution approaches π. It is common to discard an initial *burn-in* period to reduce initialization effects. Discarding a fixed number of iterations does not by itself establish convergence.

Autocorrelation. Successive MCMC iterates are dependent and nearby values are often highly correlated. Consequently, a long run can contain much less information for estimating a posterior expectation than the same number of independent samples. Efficient algorithms seek to reduce this dependence while preserving the target; Gibbs sampling is subject to this issue even though it never rejects an update.

Trace plots. Plot a coordinate or other summary of $\theta^{(t)}$ against iteration t. Rapid fluctuations around a stable region are encouraging, whereas long trends or periods of little movement suggest slow mixing. Trace plots are useful diagnostics, but a stable-looking plot alone cannot establish that the chain has explored the whole target.

Proposal choice. For random-walk Metropolis, a small proposal variance gives high acceptance but tiny steps; a large variance often proposes low-density states and gives many rejections. Both extremes slow exploration. A useful proposal balances move size against acceptance, as in the example of the preceding section.

The common principle remains the same: construct transitions preserving the posterior and use their trajectory to estimate posterior summaries. Metropolis–Hastings achieves this through an acceptance correction; Gibbs sampling uses exact conditional draws for which that correction is unnecessary.

8.4 Examples of the Gibbs Sampler

The first two examples admit direct posterior simulation, making them useful benchmarks for Gibbs sampling. Probit regression illustrates how introducing latent variables can make otherwise difficult conditional updates simple.

8.4.1 Bivariate Normal

For fixed $y = (y_1, y_2)'$, consider

$$\begin{pmatrix}\theta_1\\ \theta_2\end{pmatrix} \,\Big|\, y \sim N_2\left(\begin{pmatrix}y_1\\ y_2\end{pmatrix}, \begin{pmatrix}1 & \rho\\ \rho & 1\end{pmatrix}\right), \qquad |\rho| < 1.$$

The full conditionals are

$$\theta_1 \mid \theta_2, y \sim N\big(y_1 + \rho(\theta_2 - y_2), 1 - \rho^2\big),$$
$$\theta_2 \mid \theta_1, y \sim N\big(y_2 + \rho(\theta_1 - y_1), 1 - \rho^2\big).$$

For a random-scan sampler, initialize at $\theta^{(0)}$. At each $t = 0, \ldots, N-1$, choose $J \in \{1, 2\}$ with equal probability, sample coordinate J from its displayed conditional using the current value of the other coordinate, and leave the other unchanged.

Although every update is accepted, large $|\rho|$ causes slow mixing. The limiting case $\rho = 1$ makes the problem transparent: the target is supported on $\theta_1 - y_1 = \theta_2 - y_2$, and both conditional variances are zero. If the first update is of coordinate 1, the state becomes $(y_1 + \theta_2^{(0)} - y_2, \theta_2^{(0)})$; if it is of coordinate 2, it becomes $(\theta_1^{(0)}, y_2 + \theta_1^{(0)} - y_1)$. In either case all later updates leave that state unchanged. Even when initialized from the target, this constant trajectory cannot estimate general target expectations by time averages. The same obstruction occurs at $\rho = -1$, when the support is $\theta_1 - y_1 = -(\theta_2 - y_2)$. This example shows why stationarity and acceptance are insufficient without exploration.

Remark 8.2 (Stationarity does not imply exploration)**.** *There is no contradiction with stationarity. At $\rho = 1$, initialization from the target means drawing $Z \sim N(0, 1)$ and setting $\theta^{(0)} = (y_1 + Z, y_2 + Z)$. Every subsequent update leaves this state unchanged, so $\theta^{(t)}$ has the target distribution at every time: the chain is stationary. Along any single trajectory, however,*

$$\frac{1}{N}\sum_{t=1}^{N} \theta_1^{(t)} = y_1 + Z,$$

which does not converge to the target expectation $E_\pi[\theta_1] = y_1$. Stationarity describes the distribution at each time; ergodicity ensures that time averages along a single trajectory recover target expectations. This chain is stationary but not ergodic.

8.4.2 Linear Regression

Let $Y = (y_1, \dots, y_n)'$, with fixed design matrix X having rows x_i', and assume independent $y_i \mid \beta, \sigma \sim N(x_i'\beta, \sigma^2)$. With the improper prior $\pi(\beta, \sigma) \propto \sigma^{-1} I\{\sigma > 0\}$, the joint posterior is

$$\pi(\beta, \sigma \mid \text{data}) \propto \sigma^{-(n+1)} \exp\left\{ -\frac{\|Y - X\beta\|^2}{2\sigma^2} \right\} I\{\sigma > 0\}. \tag{21}$$

Assume X has full column rank, $n > p$, and the residual sum of squares is positive, so this posterior is proper. Direct simulation is possible by sampling a marginal posterior and then the other parameter's conditional posterior, but Gibbs sampling uses only the full conditionals.

Write $\hat{\beta} = (X'X)^{-1}X'Y$. The least-squares decomposition

$$\|Y - X\beta\|^2 = \|Y - X\hat{\beta}\|^2 + (\beta - \hat{\beta})'X'X(\beta - \hat{\beta})$$

identifies the conditional normal distribution of β. For $\tau = 1/\sigma^2$, the change of variables has absolute Jacobian $\frac{1}{2}\tau^{-3/2}$, so the conditional density of τ is proportional to $\tau^{n/2-1} \exp\{-\tau\|Y - X\beta\|^2/2\}$. Thus

$$\begin{aligned} \beta \mid \sigma, \text{data} &\sim N_p\big(\hat{\beta}, \sigma^2(X'X)^{-1}\big), \\ \tau \mid \beta, \text{data} &\sim \text{Gamma}\left(\frac{n}{2}, \frac{\|Y - X\beta\|^2}{2}\right), \end{aligned}$$

where the Gamma parameters are *shape and rate*.

Initialize at $\beta^{(0)}$ and $\sigma^{(0)} > 0$. At each $t = 0, \dots, N-1$, choose one of the two blocks with equal probability. For a coefficient update, draw $\beta^{(t+1)}$ from the normal conditional using $\sigma^{(t)}$ and keep σ fixed. For a scale update, keep β fixed, draw τ from its Gamma conditional using $\beta^{(t)}$, and set $\sigma^{(t+1)} = \tau^{-1/2}$.

8.4.3 Probit Regression: Why Introduce Latent Variables?

For binary responses and fixed covariates, probit regression assumes independent $y_i \mid \beta \sim \text{Bernoulli}(\Phi(x_i'\beta))$, where Φ is the standard normal cdf. Under a prior $\pi(\beta)$,

$$\pi(\beta \mid \text{data}) \propto \pi(\beta) \prod_{i=1}^{n} \Phi(x_i'\beta)^{y_i} \{1 - \Phi(x_i'\beta)\}^{1-y_i}.$$

With a flat prior, each coordinate conditional has this same likelihood kernel viewed as a function of β_j, with β_{-j} fixed. It is not a standard distribution from which exact simulation is immediate.

Instead, introduce conditionally independent latent variables $w_i \mid \beta \sim N(x_i'\beta, 1)$ and set $y_i = I\{w_i > 0\}$. Since $P(w_i > 0 \mid \beta) = \Phi(x_i'\beta)$, this construction gives exactly the original likelihood. **The next subsection derives simple Gibbs updates on the enlarged state space (w, β); keeping only the β draws then gives samples targeting the desired posterior**.

8.5 Gibbs Sampler for Probit Regression

Continue with the latent-variable representation above and the flat prior for β. Assume X has full column rank and the posterior is proper; a flat prior does not ensure propriety for

every binary dataset. Define $A_i = (0, \infty)$ when $y_i = 1$ and $A_i = (-\infty, 0]$ when $y_i = 0$. The augmented posterior is

$$\pi(w, \beta \mid \text{data}) \propto \exp\{-\tfrac{1}{2}\|w - X\beta\|^2\} \prod_{i=1}^{n} I\{w_i \in A_i\}.$$

Integrating out w recovers the probit posterior. The advantage of augmentation is that both block conditionals have familiar forms.

8.5.1 The Two Conditional Distributions

For fixed w consistent with the observed responses, the indicators do not depend on β. Set $\hat{\beta}_w = (X'X)^{-1}X'w$ and use

$$\|w - X\beta\|^2 = \|w - X\hat{\beta}_w\|^2 + (\beta - \hat{\beta}_w)'X'X(\beta - \hat{\beta}_w).$$

Dropping the term independent of β identifies

$$\beta \mid w, \text{data} \sim N_p\big(\hat{\beta}_w, (X'X)^{-1}\big).$$

For fixed β, the augmented density factors over i, so the w_i are conditionally independent with

$$\pi(w_i \mid \beta, \text{data}) \propto \exp\{-\tfrac{1}{2}(w_i - x_i'\beta)^2\}\, I\{w_i \in A_i\}.$$

Thus each w_i is $N(x_i'\beta, 1)$ truncated to the positive half-line if $y_i = 1$, and to the nonpositive half-line if $y_i = 0$.

8.5.2 Simulating the Truncated Normals

Fact 8.2. *If $Z \sim N(\mu, \sigma^2)$, its conditional distribution given $a < Z < b$ can be simulated as*

$$Z_{\text{trunc}} = \mu + \sigma\Phi^{-1}((1 - U)\Phi(a_0) + U\Phi(b_0)), \qquad a_0 = \frac{a - \mu}{\sigma}, \quad b_0 = \frac{b - \mu}{\sigma},$$

where $U \sim \text{Uniform}(0, 1)$, $a < b$, and either endpoint may be infinite.

To see why, let $X \sim N(0, 1)$. For $a_0 < z < b_0$, the cdf conditional on $a_0 < X < b_0$ is

$$F_{\text{trunc}}(z) = \Pr(X \le z \mid a_0 < X < b_0) = \frac{\Pr(a_0 < X \le z)}{\Pr(a_0 < X < b_0)} = \frac{\Phi(z) - \Phi(a_0)}{\Phi(b_0) - \Phi(a_0)}.$$

Inverse-cdf sampling draws $U \sim \text{Uniform}(0, 1)$ and finds the value T whose cdf equals U. Solving $F_{\text{trunc}}(T) = U$ gives

$$\Phi(T) = \Phi(a_0) + U\{\Phi(b_0) - \Phi(a_0)\}, \qquad T = \Phi^{-1}((1 - U)\Phi(a_0) + U\Phi(b_0)).$$

This has the desired distribution because $\Pr(T \le z) = \Pr(U \le F_{\text{trunc}}(z)) = F_{\text{trunc}}(z)$. Rescaling gives $Z_{\text{trunc}} = \mu + \sigma T$. Although Φ^{-1} has no elementary closed-form expression, standard numerical routines evaluate it accurately and efficiently.

With $m_i = x_i'\beta$, $\sigma = 1$, and a truncation endpoint at zero, this yields

$$w_i = m_i + \Phi^{-1}\left(\begin{cases} (1 - U_i)\Phi(-m_i) + U_i, & y_i = 1, \\ U_i\Phi(-m_i), & y_i = 0, \end{cases}\right), \qquad U_i \overset{\text{i.i.d.}}{\sim} \text{Uniform}(0, 1).$$

8.5.3 The Gibbs Algorithm

Initialize $\beta^{(0)}$ and a latent vector $w^{(0)}$ whose signs agree with y. For $t = 0, \dots, N-1$, choose $J \in \{1, 2\}$ with equal probability:

1. If $J = 1$, draw $\beta^{(t+1)} \sim N_p((X'X)^{-1}X'w^{(t)}, (X'X)^{-1})$ and set $w^{(t+1)} = w^{(t)}$.
2. If $J = 2$, set $\beta^{(t+1)} = \beta^{(t)}$ and independently draw each $w_i^{(t+1)}$ using the truncated-normal formula above with $m_i = x_i'\beta^{(t)}$ and fresh independent uniforms.

One may instead update both blocks sequentially in every sweep, using the newly drawn block in the next update. After burn-in, use the β iterates for posterior summaries. The latent variables simplify simulation without changing the marginal posterior of interest.